

Dataset	Queries	Items	Rel.	Feats.
Airline Itinerary	33951	1,089k	2	17
Microsoft 30k	31531	3,771k	5	136
Istella-S	33018	3,302k	5	220

Table 1: Characteristics of the datasets used in the experiments: number of queries, items, relevance levels, and features.

and use the standard discounted cumulative gain (NDCG) metrics that include NDCG@1, NDCG@3, NDCG@5, and NDCG@10. For Microsoft 30k dataset, an additional metric NDCG@50 was also evaluated. On the other hand, for user choice ranking tasks where relevance feedbacks are binary-valued, we follow (?) and use top precision metrics that include P@1(Precision@1) and P@5.

Model Training

Each dataset is split in train, validation and test sets according to a 60%-20%-20% scheme. The validation set was used to select the optimal hyperparameters for all involved methods. We did not perform extensive hyperparameter search for the proposed model, and used virtually the same architecture throughout all the experiments and datasets. More specifically, the dimensions of the nonlinear transformations (1) in the Input Encoder were fixed as 100, while MLPs used in (3) and (9) consist of 2 hidden layers with either 256 or 128 ELUs. The models were trained with the Adam algorithm (?) with a learning rate of 0.001, batch size of 80. Training generally converged after less than 100 passes through the entire training dataset.

Comparisons with Baseline Methods

The performance comparison results of various methods are reported in Table 2. To eliminate the influence of random initiations, all results are averaged over 20 runs. As is shown in this table, QILCM significantly outperforms all the baselines.

Ablation Studies

To elucidate the contributions of the main components of our system, in this section, we test several variants of the proposed model. The tested implementations include:

- **Variant 1:** Our model with the attention-weighted context encoding (4) replaced by simple average pooling of the item encoding vectors. Accordingly, the attention-weighted statistics (6) and (7) are replaced with standard mean and variance;
- **Variant 2:** Our model without the domain confusion loss;
- **Variant 3:** Our model without the domain confusion loss and QN layer.

The performance results of different implementations of the proposed methods are shown in Table 3, which shows that QILCM consistently achieves the best performance among all model variants.

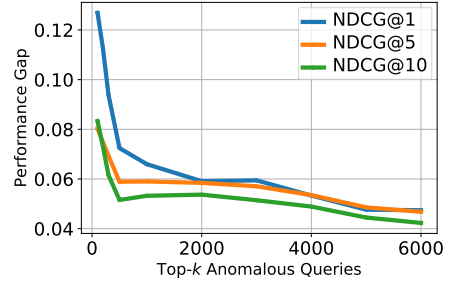


Figure 2: Performance gap between QILCM and the best-performing baseline (DLCM) on Microsoft 30k dataset.

Anomalous Query Analysis

To further investigate the performance of QILCM, in this section we conducted additional analysis of experimental results on the Microsoft 30k dataset. Concretely, we firstly followed (?), and constructed vector representation for each query by averaging over the feature values of the top k ranked items (k was set as 10 during the experiments). After that, we fitted Isolation Forest (?) to the training queries, and then used the learned model to assign a ‘anomaly score’ to each query in the test set. Intuitively, this score quantifies how different a query is from the majority of training data, and we examine the performance difference between QILCM and the best-performing baseline (DLCM) for queries with different levels of anomaly. As shown in Figure 2, the performance gap between QILCM and DLCM is significantly widened for more anomalous queries. For example, the NDCG@1 gap between QILCM and DLCM is increased from 0.047 to 0.121, which clearly demonstrates the advantage of tackling heterogeneous queries using the proposed DG perspective.

Conclusion

In this paper, we introduce a DG formulation of the LETOR problem and propose a novel neural architecture for DG in this LETOR context. We evaluate our techniques on three benchmark datasets, demonstrating that the proposed approach outperforms previous state-of-the-art approaches by a substantial margin.

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Dataset	Metrics	QILCM	DCM	DLCM	LambdaMART	Improv.	P-value
Airline Itinerary	P@1	*0.2833	<u>0.2618</u>	0.2562	0.2327	8.21%	2.40×10^{-23}
	P@5	*0.6958	<u>0.6586</u>	<u>0.6651</u>	0.6239	4.61%	1.22×10^{-20}
Microsoft 30k	NDCG@1	*0.5447	0.4938	<u>0.4973</u>	0.4800	9.53%	3.83×10^{-19}
	NDCG@3	*0.5313	<u>0.4827</u>	0.4811	0.4766	10.06%	7.58×10^{-16}
	NDCG@5	*0.5368	<u>0.4904</u>	0.4892	0.4842	9.46%	1.60×10^{-16}
	NDCG@10	*0.5564	0.5093	<u>0.5135</u>	0.5061	8.35%	2.27×10^{-15}
	NDCG@50	*0.6482	0.6127	<u>0.6146</u>	0.6092	5.46%	4.51×10^{-15}
Istella-S	NDCG@1	*0.7023	0.6762	<u>0.6873</u>	0.6644	2.18%	1.74×10^{-10}
	NDCG@3	*0.6696	<u>0.6552</u>	0.6537	0.6378	2.20%	1.99×10^{-8}
	NDCG@5	*0.6953	<u>0.6846</u>	0.6831	0.6741	1.56%	4.25×10^{-10}
	NDCG@10	*0.7645	0.7558	<u>0.7566</u>	0.7456	1.04%	4.67×10^{-7}

Table 2: Performance comparison of various methods. The results are averaged over 20 random runs, and the best ones are marked with *. The last two columns show the improvement of QILCM over the best baseline algorithm (highlighted with underline), and the corresponding Student’s t-test P-values.

Dataset	Metrics	QILCM	Variant 1	Variant 2	Variant 3
Airline Itinerary	P@1	*0.2833	0.2749	0.2762	0.2694
	P@5	*0.6958	0.6613	0.6803	0.6724
Microsoft 30k	NDCG@1	*0.5447	0.5287	0.5359	0.5139
	NDCG@3	*0.5313	0.5127	0.5294	0.4970
	NDCG@5	*0.5368	0.5230	0.5347	0.5030
	NDCG@10	*0.5564	0.5459	0.5538	0.5229
	NDCG@50	*0.6482	0.6363	0.6438	0.6331
Istella-S	NDCG@1	*0.7023	0.6966	0.6982	0.6837
	NDCG@3	*0.6696	0.6654	0.6672	0.6628
	NDCG@5	*0.6953	0.6936	0.6931	0.6923
	NDCG@10	*0.7645	0.7622	0.7637	0.7602

Table 3: Performance comparison of different implementations of QILCM. The results are averaged over 20 random runs, and the best ones are marked with *.

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